

FREIGHT FORWARDING & LOGISTICS ERP

Documentation

FreightFast is an enterprise-grade freight forwarding ERP — shipment requests, quotations, job files, freight documents, finance closeout, and a client portal, in one system. This page covers what it is and everything it does. For setup steps, see the Installation Guide.

FRAMEWORK Next.js 16 (App Router)	LANGUAGE TypeScript, React 19	DATABASE MySQL / MariaDB + Prisma	AUTH NextAuth.js	DOCUMENTS PDFKit
AI LAYER Google Gemini (optional)				

01 — READ FIRST

What FreightFast is

A single Next.js application with a MySQL-compatible database. There's no separate backend service to stand up — the app, API routes, and background job endpoints all ship from the same codebase.

Three surfaces, one app

The build serves three distinct sign-in surfaces from the same deployment: an **operations dashboard** for company staff (quotations, shipments, finance), a **platform console** for the SaaS operator managing tenant companies, and a **client portal** your customers use to track their own shipments and documents. Routing between them is host-based, configured during installation.

Multi-tenant by design

Every record belongs to exactly one company, and within a company, to exactly one branch. If you're installing this for a single freight forwarder, you'll use one company with one or more branches; the platform console exists for cases where you're operating this as a SaaS across multiple client companies.

Role-based from the ground up

Seven predefined roles — Company Admin, Operations Manager, Documentation Officer, Accounts Officer, Sales Executive, and two more — gate every module, action, and record. Permissions are enforced server-side on every request, not just hidden in the UI.

02 — WHAT'S INSIDE

Core features

Eighteen operational modules covering the full shipment lifecycle, from first inquiry to closed invoice.

Control Tower Dashboard

Real-time KPIs, shipment visibility, workflow blocker detection, revenue & profit analytics.

Customer & CRM Management

Customer database, company profiles, contacts, billing details, shipment history.

Shipment Request Management

Inquiry intake, cargo capture, Port-to-Port / Door-to-Door scope, auto-filled ETD/ETA.

Quotation Management

Freight & local charge estimation, buy/sell rates, profit visibility, approval workflow.

Job File Management

10-tab workspace: Overview, Workflow, Commercial, Operations, Documents, Freight Docs, Finance, Portal, Tasks, Timeline.

Document Compliance Engine

Mode/type-aware document requirements, required-document gates, finance close blockers.

Freight Document Generator

PDF generation for HBL, HAWB, sea/air manifests, delivery orders, debit notes, arrival notices.

Finance & Closeout

Invoicing, vendor bills, margin audit, financial lock, audit-safe closeouts.

Client Portal

Role-Based Access Control

OTP login, forced first-time password reset, milestone tracking, document access.

Seven predefined roles, fine-grained permissions, strict company isolation.

Multi-Branch Management
Every record owned by exactly one branch; staff scoped to the branches they belong to.

Automated Status Notifications
In-app alerts on invoice/payment/POD events, plus an idempotent email/WhatsApp outbox.

Advanced Global Search
One search box across customers, shipments, quotations, invoices, vendors, and documents.

Saved Filters & Views
Private, named filter presets on list pages, with a per-user default view.

Customizable Dashboard
Reorder and show/hide dashboard sections per user, with a one-click reset.

Import & Export Data
CSV export across eight entities; a guided import wizard for customers and vendors.

Recurring Task Management
Daily/weekly/monthly task templates, generated by an idempotent self-hosted cron.

Custom Document Templates
Branded, versioned layouts for quotations, invoices, HBL, HAWB, debit notes, manifests.

Approval Workflow System
Threshold-based sign-off chains for vendor bills and payments, up to five approver steps.

03 — OPTIONAL LAYER

AI-assisted tools

Gated behind a platform kill-switch, a per-company toggle, and a daily request cap — off by default until a Gemini API key is configured. See the Installation Guide's environment reference for `AI_ENABLED`.

TOOL	WHAT IT DOES	POWERED BY
Shipment Assistant	Chat-style Q&A over a shipment's status, tasks, and open issues.	Gemini

Document Reader	Extracts shipment and document fields from an uploaded file.	Gemini
Document Checker	Flags missing, incorrect, or inconsistent fields on a freight document.	Gemini
Document Cross-Check	Compares multiple documents on a shipment and flags mismatches.	Gemini
Quotation Generator	Drafts charge lines from customer, shipment, and pricing context.	Gemini
Email Generator	Drafts a customer-ready email from shipment and operational context.	Gemini
Shipment Summary	Turns a full job file into a short operational summary.	Gemini
Report Insights	Explains notable trends and changes on financial reports.	Gemini
Smart Search	Natural-language mode for the global search box.	Gemini
Customer Insights	Summarizes a customer's history, activity, and opportunities.	Gemini
Delay Alerts	Flags shipments at risk of delay from ETA and workflow data.	Rule-based
Profit Analysis	Highlights low-margin shipments against historical averages.	Rule-based
Task Suggestions	Recommends next actions from open exceptions and pending work.	Rule-based
Shipment Health Score	Scores a shipment from status, milestones, documents, and conditions.	Rule-based
Exception Radar	Surfaces shipments, documents, or workflows that need attention.	Rule-based aggregator

 "Rule-based" tools don't call an external model — they're deterministic logic over your own data. They still route through the same permission and company-toggle gate as the Gemini-backed tools, so access control stays consistent either way.

Is your hosting compatible?

Ask your host, or check your control panel, for these three things.

YOU NEED	TYPICAL SHARED CPANEL HOSTING	VPS / CLOUD SERVER
SSH / terminal access	usually no	yes
Ability to run a persistent Node.js process	usually no	yes
Your own MySQL/MariaDB database with remote or local access	often yes	yes

 If your plan is marketed as "shared hosting for WordPress/PHP sites" and doesn't mention SSH access, it will not run this application. This is the single most common cause of a failed install — check this first, before buying.

05 — SIZING

Server specs

Actual needs scale with company size and concurrent users. These cover a small-to-mid-size deployment.

RESOURCE	MINIMUM	RECOMMENDED
CPU	1 vCPU	2 vCPU
RAM	1 GB	2 GB+
Disk	10 GB	20 GB+ (more if storing documents)
OS	Any Linux distro that runs Node 20+	Ubuntu 22.04/24.04 LTS
Node.js	20.x	20.x LTS or newer
Database	MySQL 8.0 / MariaDB 10.6, same server or managed	Managed database service for p

06 — CONFIRMED TO WORK

Where it runs

Anywhere you get a real Linux server with SSH access and can run `npm run build && npm run start` as a long-lived process.

Any VPS provider

AWS EC2 / Lightsail

DigitalOcean, Linode, Vultr, Hetzner, Contabo — a basic droplet/instance is enough to start.

Full control over the instance; pair with RDS for a managed database if desired.

Azure App Service

A ready-made GitHub Actions workflow for this is already included in the repository.

Self-managed VPS over SSH

A ready-made GitHub Actions workflow for SSH-based deploys is already included in the repository.

Docker host

Any server or platform that runs a Dockerfile you build from this codebase (Next.js standalone output works well in a container).

Serverless platforms (Vercel, Netlify, and similar): possible, but not the primary supported path. They require extra configuration for the database connection pooling Prisma needs under serverless invocation, and for scheduling the two background-job endpoints (which need a real cron trigger, not just an HTTP route). The two deploy workflows bundled with this codebase target a VPS and Azure App Service specifically because both give a persistent process and a normal system cron.

07 — KEEPING IT RUNNING

Process management & scheduled jobs

A process manager, not just `npm run start`

Running `npm run start` directly in a terminal stops the moment that terminal closes. In production, run it under a process manager that restarts it automatically — `pm2`, a `systemd` service, or a Docker container with a restart policy all work.

A reverse proxy in front of it

Next.js serves plain HTTP on its own. Put Nginx (or equivalent) in front of it for TLS/SSL termination and to serve on ports 80/443.

Cron access for two optional features

Automated external notification delivery and recurring-task generation both rely on something calling a protected URL on a schedule — a normal Linux `crontab` entry, or any external scheduler that can make an HTTPS request. Both features work perfectly well without this

configured; they just don't run on autopilot until it is. See the Installation Guide's Background Jobs section for the exact commands.

08 — WON'T WORK

What won't work

HOSTING TYPE	WHY NOT
Shared cPanel/WordPress hosting without SSH	No way to install Node.js dependencies or keep a Node process running.
Static file hosting (e.g. plain S3 bucket, GitHub Pages)	This is a server-rendered application with API routes and a database — not a static site.
PHP-only hosting	There's no PHP in this codebase to run; the host needs a Node.js runtime.
Any host without database access	A MySQL/MariaDB database is a hard requirement — either on the same server or reachable over the network.

09 — AT A GLANCE

What needs an account, and what doesn't

FreightFast can integrate with a handful of external services. Exactly one is required to run the app at all — the rest are optional, and each feature that needs one simply stays switched off until it's configured.

SERVICE	USED FOR		WHERE TO GET IT
MySQL / MariaDB database	All application data	required	Self-hosted on your server, or a managed provider
Azure Blob Storage	Uploaded shipment documents	optional	portal.azure.com
Google Gemini (Google AI Studio)	10 of the 15 AI-assisted tools	optional	aistudio.google.com
SMTP provider	Outbound email — notifications, client portal credentials	optional	Any SMTP relay
WhatsApp Cloud API (Meta)	WhatsApp notification delivery	optional	developers.facebook.com

 Everything below the database is opt-in. Install the app, run it, and add these later as you need them — nothing has to be decided on day one.

10 — REQUIRED

Database

The only hard external dependency. Every record — shipments, quotations, invoices, users — lives here.

Self-hosted: install MySQL 8.0 or MariaDB 10.6+ on the same server as the app. No ongoing cost beyond the server itself.

Managed: a hosted MySQL-compatible database (e.g. Amazon RDS, DigitalOcean Managed Databases, PlanetScale) removes the need to administer the database server yourself, at the cost of the provider's hosting fee. Either path works — set `DATABASE_URL` to point at whichever you choose.

11 — OPTIONAL

File storage — Azure Blob Storage

Powers the Shipment Document Vault — uploaded freight documents, HBL/HAWB copies, and other shipment files.

Without `AZURE_STORAGE_CONNECTION_STRING` configured, document upload is unavailable but the rest of the app functions normally. Azure's free-tier credits for new accounts are typically enough to evaluate the feature before committing to a paid plan.

12 — OPTIONAL

AI features — Google Gemini

Ten of the fifteen AI-assisted tools (Document Reader, Document Checker, Quotation Generator, Email Generator, and others) call the Gemini API. The remaining five are deterministic and need no external service.

Get a key from Google AI Studio and set `GOOGLE_GENERATIVE_AI_API_KEY`, plus `AI_ENABLED=true`. Gemini's free tier covers moderate usage; heavier volume moves to its paid pricing. Without a key configured, every AI feature — including the five deterministic ones, which share the same permission gate — stays switched off.

Delivery channels

In-app notifications always work with no setup. These three only govern sending a copy externally.

Email — any SMTP provider

Point `SMTP_HOST` / `SMTP_PORT` / `SMTP_USER` / `SMTP_PASS` at any SMTP relay — a transactional email provider, your own mail server, or even a personal account for testing. Pricing and free-tier limits vary by provider.

WhatsApp — Meta WhatsApp Cloud API

Requires a Meta for Developers app with the WhatsApp Business Platform enabled. Meta bills per conversation once you're past its free tier — see Meta's own pricing for current rates.

SMS — any SMS gateway

Plug in whichever regional SMS API provider you prefer via `SMS_PROVIDER` / `SMS_API_KEY`. Cost is per-message and varies significantly by provider and destination country.

Support

For setup and configuration issues, start with the Installation Guide — it covers environment variables, default accounts, and a troubleshooting section for the most common setup problems. Keep both documents alongside your copy of the source for reference during future upgrades.